

When Self-Modification Becomes Memory

An Audited Comparison of Three Self-Improving Agent Substrates

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Abstract

Language-model agents can edit prompts, memories, tools, and code, but self-improvement bundles five evidentiary propositions: retention, expression, behavioral mediation, task improvement, and recursive improvement. This study instruments the first four; the fifth requires descendant-productivity evidence. We audit three substrate-proposal-acceptance-interface configurations built around a workspace, a procedure store, and an executable Pack. They share an outer agent, a 28-task development curriculum, and cold-restart evaluation on 19 held-out tasks. None of the three recorded lineages met the frozen uplift criterion. The shared interface expressed retained structure mainly as text; per-task item coverage ranged from 12% to 100%, and the actor had no retained invocation handle. Six actor-visible-equivalent no-context arms issued byte-identical initial requests per task yet differed on 7 of 19 tasks. A code audit found that the nominal state-hidden ablation was not separable from cold at the actor-visible boundary, so we retain its contrast as the frozen decision rule and add pooled-six-control sensitivity; the null is unchanged. Across nine configuration-family cells, an opaque sham scored below the labeled ablation in five, above it in three, and tied once, so the design cannot isolate possible sham interference from outcome spread. Mechanism audits found policy-cap saturation, a common proposal-form bottleneck, and vocabulary-associated selection of failure-derived lessons. Two separately executed, outcome-blind model-coding passes agreed that all 84 proposals shared one nonexecutable textual genre; one coder also generated the proposals, and interpretive labels remain exploratory. We contribute an expression profile, an equivalent duplicates protocol, and a control ladder. In this harness, the interface and acceptance mechanism are measurable components of the self-improving system.

1 Introduction

Language-model agents now edit their own prompts, memories, tools, and source code, and the reported results are striking. STOP [1] recursively improves its own improver. SICA [2] raises its score on a SWE-bench Verified subset from 17% to 53%. The Darwin Gödel Machine [3] more than doubles SWE-bench performance across a branching archive of self-modified descendants, and the Huxley-Gödel Machine [4] scores lineages by the productivity of their descendants. These systems make self-modification

easy to observe. Self-improvement is a different empirical claim. A visible edit establishes that the system changed itself. It does not establish that the change was expressed at evaluation time, altered behavior, caused better externally graded outcomes, or improved the system’s capacity to produce useful future changes.

We separate the statement that an agent improved itself into five constructs: (1) **retention**, meaning persistent state changed; (2) **expression**, meaning retained state reached the evaluation-time policy; (3) **behavioral mediation**, meaning the policy’s trajectory or resource use changed; (4) **task improvement**, meaning externally graded held-out performance rose beyond matched controls and stochastic spread; and (5) **recursive improvement**. These are distinct evidentiary propositions, not ordinal performance levels. Retention and actor-visible expression can be established descriptively; behavioral mediation, task improvement, and recursive improvement generally require counterfactual or lineage contrasts. This study instruments the first four and frames descendant productivity as the required operationalization of the fifth.

We study those links in a heavily instrumented comparison of three persistent self-modification substrates chosen for the breadth of their native update surfaces: an arbitrary rewritable workspace, a success-gated procedure store, and a governed, hash-pinned executable ActiveGraph Pack. Each traversed the same ordered 28-task development curriculum and then faced 19 disjoint held-out tasks across three families after a cold restart. The design used external graders, four arms, immutable receipts, hash-bound adjudication, and a frozen success criterion. It also had one development lineage per configuration, one model, and small task families. The orchestration and local ActiveGraph family were self-built; SWE and Terminal used external official graders, and every claim is scoped to the recorded harness. The frozen result is null: none of the three recorded lineages met the criterion. This paper explains why that null is informative and why several favorable and unfavorable results inside our data dissolve under audit.

The audit measures mechanisms that recur across agent systems. First, budgeted context is a common route from memory to later action. Under that route, our three substrate-interface configurations delivered between 12% and 100% of their declared retained items per task, exposed no retained invocation handle in the actor-visible action schema, and expressed native structural change to the acting policy almost entirely as text. These per-

centages are within-substrate coverage measures rather than comparable quantities of useful information. Second, single-run task estimates can conceal substantial outcome variability. Our six no-context labels issued byte-identical first requests per task, seed included, and still differed on 7 of 19 tasks. Their family-level means spanned 2/10 tasks (20.0 points) on SWE and 1/6 task (16.7 points) on Terminal; the continuous-score ActiveGraph span was 21.3 points. Third, placebo context can change the computation it is meant to control. On SWE, the Workspace evolved arm solved 8/10 tasks and the sham solved 6/10, creating a 20-point evolved-minus-sham contrast; the six equivalent no-context labels on the same ten tasks already ranged from 6/10 to 8/10. Fourth, lexical retrieval and append-only retention created observable selection and capacity effects. A top-three retriever over-selected broad-vocabulary failure lessons, and the Pack eventually blocked three consecutive updates at its policy limit.

The shared reflection layer created a proposal-form bottleneck before configuration-specific acceptance and native mutation. Two separately executed outcome-blind model-coding passes using two model families over all 84 update proposals assigned every proposal to the same representational genre: natural-language, append-only, nonexecutable, operational guidance with no specified activation. Interpretive fields were less stable across coders and remain exploratory pending human coding. One coder, Sol, was also the model that generated the proposals; Terra supplied the different-family pass. The cross-model consistency finding concerns the proposal layer. Native Workspace edits and Pack source changes remained structurally real and heterogeneous.

The receipts cut in both directions. An apparent transfer success involved an evolved SymPy pass matched by a no-context run that produced essentially the same fix, while the retained SymPy lesson concerned a different mechanism. An apparent retrieval failure scored exactly at the minimum of six equivalent no-context runs. The one evolved result outside its task’s exact equivalent range was a regression in which retained context accompanied a larger invalid intervention and the ablation produced the smaller correct fix. A tie-aware conditional exchangeability calculation expected 12/7 (1.71) outside-range results and observed one, so the case is a mechanistic trace rather than a population or chance detection. A case audit that can reject favorable and unfavorable causal stories symmetrically is a minimum requirement for interpreting individual trajectories.

Prior systems provide strong evidence for the forms of optimization they measure. Table 6 records which studies report held-out transfer, mechanism ablations, sham retained state, independent evolution runs, byte-identical initial-request duplicates, immutable receipts, and descendant productivity. We do not reinterpret their reported gains through our experiment. Under our controls, our apparent gains dissolved. Applying the same instruments in independent systems is a concrete next test.

This paper contributes six reusable instruments and ob-

servations:

1. a five-construct decomposition separating retention, expression, behavioral mediation, task improvement, and recursive improvement;
2. an **expression profile** measuring storage reach, eligible-payload reach, semantic-unit reach, selection adaptivity, and execution reach;
3. an **equivalent duplicates** protocol that creates task-level empirical dispersion references from executions with byte-identical initial requests;
4. a **control ladder** organizing eight control designs by the nuisance they address and the claims they license;
5. mechanism checks for **failure-memory stickiness**, capacity saturation, and common-template reflection-proposal homogenization; and
6. a receipt-bound compact release with graders, requests, artifacts, and an adjudication ledger; complete trajectory traces remain in a separately gated archive pending disclosure review and deposit.

The scope is a within-study diagnostic demonstration. It does not supply a population estimate, a universal architecture ranking, or a universal auditability-expressivity law. The expression bottleneck was created by the implemented common interface, and that fact is central to the result. The reflection template determined what could be proposed, the acceptance gate determined what survived, the retriever and adapter determined what was expressed, and the execution stack determined what a score could mean. The interface and the acceptance mechanism are part of the self-improving system.

2 Measurement framework

Let R_i^t denote retained state for configuration i after development generation t . Let $A_i(R_i^t, x)$ be the evaluation adapter for task x , $O_i^t(x)$ the observation and action surface emitted by the adapter, $\pi(O_i^t(x), x)$ the induced acting policy, $Y_i^t(x)$ the external task score, and M_i^t the productivity of the process that creates useful descendants. The five constructs correspond to different comparisons:

1. Retention: $R_i^{t+1} \neq R_i^t$.
2. Expression: $O_i^{t+1}(x) \neq O_i^t(x)$.
3. Behavioral mediation: the induced policy trajectory changes.
4. Task improvement: expected external score increases beyond matched controls and relevant outcome spread.
5. Recursive improvement: the productivity M_i^t of generating useful descendants increases.

This decomposition prevents evidence from moving silently between levels. The five propositions have different logical status and need not move together. A new file establishes retention. A serialized lesson establishes possible expression. A changed trace establishes behavioral mediation. An external score contrast supports task improvement. Recursive improvement is the construct; descendant productivity is its operationalization. It requires evidence that earlier changes improved later update pro-

Figure 1. The expression bottleneck

Persistent change is filtered before it can affect action; each arrow is a distinct empirical claim.

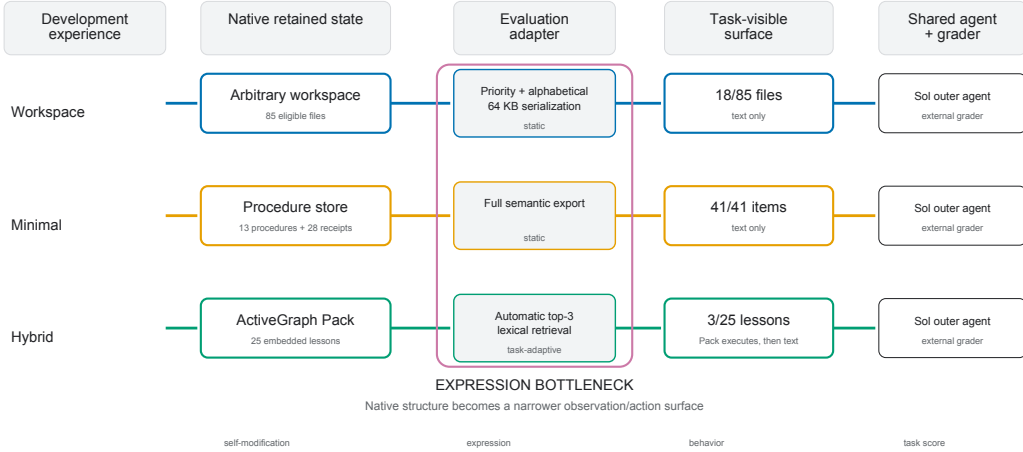


Figure 1: The expression bottleneck. The study separates retention, expression, behavioral mediation, task improvement, and recursive improvement. Retention is necessary but does not identify later score improvement or recursive self-improvement. The bold bottleneck is the implemented adapter boundary. It does not imply that all auditable interfaces reduce expressivity.

duction.

2.1 Expression profile

A single byte ratio cannot compare heterogeneous stores. Minimal’s complete SQLite state is mostly provenance, while its semantic export is small. Workspace bytes mix source, tests, instructions, and records. Hybrid executes a Pack query before the actor, so direct byte delivery omits runtime activation. We therefore report five structured axes:

1. **storage reach**: unique bytes copied into the initial actor-visible context divided by bytes in a declared canonical serialization of the complete frozen retained-state snapshot;
2. **eligible-payload reach**: those copied bytes divided by the canonical retained byte spans admitted by the adapter’s frozen eligibility rule;
3. **semantic-unit reach**: declared substrate-native retained items surfaced per task, plus their union across the probe, divided by the frozen item inventory;
4. **selection adaptivity**: whether selection is explicitly conditioned on task input, accompanied by the per-task-to-probe-union coverage expansion; and
5. **execution reach**: whether retained code runs automatically and how many retained invocation handles appear in the actor-visible action schema.

Semantic units are files for Workspace, typed procedures, capabilities, or receipts for Minimal, and lessons for Hybrid. The unitization rule and item types are frozen before evaluation. A Workspace file counts once when any bounded prefix of that file reaches the actor; completeness is reported separately. These ratios measure coverage within one substrate and must not be used to compare quantities of useful information across substrates.

The coverage-expansion gap measures selection turnover; it does not by itself show that selection is relevant or beneficial. Interfaces without byte-copy semantics should report the byte axes as not applicable and declare a proxy rather than force a ratio.

The expression profile is diagnostic rather than ordinal. Greater reach can increase useful transfer, interference, cost, or all three.

2.2 Control ladder

The control ladder organizes controls by the nuisance they address and the inference they license. Rungs 3 through 6 are alternative payload controls, not a monotonic sequence in which every higher number dominates every lower number.

The present study separably implements rungs 1, 3, and 7. It records the lineage state hash required by rung 2, but the cold-ablation adapters validate and then suppress that state without exposing a lineage-derived channel to the actor. Cold and cold ablation therefore collapse at the actor-visible intervention boundary. Rung 2 is not separably implemented here. The other rows are operational specifications for follow-up designs rather than empirical controls claimed here.

3 Study design

All three configurations used the same `gpt-5.6-sol` outer coding agent, brokered tool protocol, task ceilings, grader boundary, and common reflection step. Accordingly, the experiment compares three proposal, acceptance, retention, and interface configurations downstream of one shared textual proposal generator. It does not compare unconstrained native self-modification processes. The

Control ladder. Payload controls 3–6 are alternatives rather than a monotonic sequence.

Control	Intervention and held-fixed variables	Licensed inference
1. Cold floor	Same model, task, harness, and budget; no lineage	Model-and-harness floor
2. State-hidden ablation	Same developed lineage and evaluation stack; learned state hidden; a distinct actor-visible channel is required	Effect of enabling retained state only when that distinct channel is verified
3. Opaque byte-size sham	Seeded opaque text; report byte ratio and token load	Sensitivity to opaque payload of recorded size
4. Token-matched neutral sham	Match tokenizer, position, and count with a preregistered corpus and blinded relevance screen	Payload effect after equalizing first-call token load
5. Semantic sham	Plausible units from disjoint task families under blinded relevance screening	Sensitivity to screened-unrelated guidance
6. Structure-preserving sham	Preserve count, schema, and token budget while permuting content or metadata	Sensitivity to retained structure apart from its original mapping
7. Equivalent duplicates	Byte-identical initial request, model, seed field, task image, harness, and budgets; later trajectories may diverge	Observed within-task execution dispersion
8. Native-interface ablation	Hold state, task, model, and budgets fixed while toggling native execution	Effect of the native execution path with recorded downstream changes

configurations differed in retained state:

- **Workspace v1.2** could rewrite and retain an arbitrary workspace tree.
- **Minimal v2** stored every scalar receipt and promoted only successful experiences as procedures or deterministic capabilities.
- **Hybrid Packs** used a model-authored, tested, hash-pinned ActiveGraph Pack to store and retrieve lessons.

The native mutation units differ in breadth. The score comparison concerns their implemented common interface and does not order their native ceilings.

Each configuration traversed the same 28-task development curriculum: 20 SWE-bench Verified issues, six Terminal-Bench 2 tasks, and two local Ouro-ActiveGraph-50 systems. Evaluation used 19 disjoint held-out tasks: ten SWE-bench Verified issues, six Terminal-Bench 2 tasks, and three Ouro-ActiveGraph-50 systems. SWE and Terminal used official binary verifiers. The ActiveGraph family used normalized scores from 50 sealed checks. The local benchmark and the Hybrid Pack both use ActiveGraph technology, creating a structural author-built overlap. The sealed checks and the unfavorable Hybrid ActiveGraph result, -10.0 points versus its labeled ablation, mitigate simple directional bias but do not remove that conflict.

For each configuration and held-out task, the study ran four arms:

- **evolved**, with retained development state enabled;
- **cold ablation**, with the same lineage hash validated and learned state hidden;
- **sham**, with a seeded opaque artifact intended to approximate the evolved context size; and
- **cold**, with no development state.

Source inspection after Round 1 confirmed that cold and cold ablation have the same task, model, seed field, tools, workspace, system instructions, empty retained-context field, and later actor-visible tool protocol. The hidden lineage hash is validated for cold ablation but never reaches the actor. The frozen primary estimand, paired **evolved** – **cold ablation**, is therefore an evolved contrast against one labeled draw from the six actor-visible-equivalent no-context arms rather than a separable matched-lineage intervention. We keep that con-

trast as the frozen decision rule and report pooled-six and leave-the-matched-ablation-out sensitivities in Section 4.

The frozen rule required benefit over ablation and sham, a nominal 95% task-resampling interval above zero in at least two families including SWE or Terminal, valid cold reload, and no material development regression. This was an operational decision rule for the recorded probe rather than a claim of population-level interval coverage.

The information boundary excluded hidden tests, expected outputs, private cases, grader logs, and oracle artifacts from lineage state. Each accepted generation produced an immutable directory whose manifest bound architecture, parent run, development experience, artifact size, and hash. Evaluation occurred after cold reload. The study specification is `../research/local_study.json`, SHA-256 `8313d21971452704893200d2022718fec985142ae32fe7d16932d34a0e0d581`, committed with frozen status at study commit `559843894578141dfbd39b7bae2246db76e57bde` on 2026-07-17T18:30:38-07:00. The compact release does not preserve an independently witnessed timestamp for the first development call, so we call this a frozen study specification and decision rule rather than an externally registered preregistration. The manuscript package reviewed in Round 1 was commit `8bdf5f02cc08be968b21bbe5ebf177d49879a0fb`.

The study completed 84 development attempts and 228 held-out attempts. Failed attempts remained in the denominator. Six SWE runs reached the frozen 3,600-second task limit. Six Terminal agents reached the 1,800-second limit, and completed verifiers returned zero. One ActiveGraph grader contradicted a receipt-bound parsable submission; one sealed grader-only retry of the hash-identical artifact recovered 24/50. No model attempt was rerun and no raw row was rewritten. The adjudication index binds the raw report, datasets, changed analytical row, and evidence by SHA-256.

Scores remain separate by family. Nominal 95% descriptive task-resampling intervals use 10,000 deterministic percentile-bootstrap draws. Each draw resamples the paired per-task arm differences jointly with replacement. With one lineage and ten, six, and three held-out tasks, no frequentist coverage is asserted. The three-task Active-

Graph interval is especially discrete and unstable.

4 Frozen outcome rule and pooled-control sensitivity

None of the three recorded lineages met the frozen uplift criterion. Table 1 reports the binary families as fractions and percentages. For context, the six equivalent no-context labels ranged from 6/10 to 8/10 on SWE and from 4/6 to 5/6 on Terminal despite byte-identical initial requests within each task.

Intervals equal to $[0, 0]$ are degenerate because every paired task difference was zero; they do not imply population-level certainty. The labeled ablation column preserves the frozen comparison even though source inspection later showed that it was one no-context draw rather than a separable lineage intervention.

The pooled-six sensitivity recomputes the descriptive mean contrast against all actor-visible-equivalent no-context labels. In Workspace, Minimal, and Hybrid order, the evolved-minus-pooled deltas are +6.7, -3.3, and +6.7 points for SWE; +11.1, -22.2, and -5.6 points for Terminal; and -9.9, +4.1, and -9.9 points for ActiveGraph. Leaving each configuration’s labeled ablation out of its reference gives +8.0, -6.0, and +6.0; +10.0, -20.0, and -6.7; and -8.4, +3.2, and -9.9 points, respectively. None changes the frozen no-uplift decision. The exact nine-cell calculation is `data/generated/equivalent-control-sensitivity.csv`.

The table and sensitivity establish persistent evaluation after development and the absence of frozen uplift. They do not explain whether retained state was available, whether it changed behavior, or whether visible contrasts exceeded outcome variability. Sections 5 through 7 audit those links.

5 Expression bottleneck

All three substrates created cold-loadable state:

These receipts establish retention. The expression profile shows what could reach later behavior:

Minimal supplies the decisive counterexample to storage reach as a sufficient metric. Its database and provenance artifacts were large, giving 1.35% storage reach, while the adapter exported all declared semantic units. Those 41 units comprised 13 procedures and 28 receipts. None was a deterministic capability. The byte numerators and denominators are Workspace 64,000/429,731 storage and 64,000/247,411 eligible payload; Minimal 24,767/1,839,823 and 24,767/24,865; and Hybrid mean 10,677/266,265 and 10,677/62,319.

Workspace exposed 17 complete files and a bounded prefix of an eighteenth. Under the frozen semantic-unit rule, a file counts once when any bounded prefix reaches the actor, yielding 18/85; completeness is reported separately. All post-priority files came from the alphabetically

early `_development/` directory, and the actor could not invoke retained tools. Minimal delivered the same complete export to every task. Hybrid alone selected different learned units by task: three of 25 per task and 17 of 25 across the probe. Its Pack executed a retrieval before each task, while the actor received only the returned text and could not call the Pack interactively.

The common adapter therefore narrowed each native substrate in a different way. Workspace was truncated and static. Minimal was semantically complete and static. Hybrid used task-conditioned selection with 56 points of probe-union coverage expansion. No actor-visible action schema contained a retained invocation handle. Minimal had no retained executable unit, while Workspace and Hybrid retained executable material that the actor could not invoke on demand. This supports an expression bottleneck in the implemented interface. A general auditability-expressivity tradeoff remains a hypothesis for a future interface intervention.

6 Equivalent duplicates and control interference

The six no-context labels were Workspace cold, Workspace ablation, Minimal cold, Minimal ablation, Hybrid cold, and Hybrid ablation. For each task, their first model requests were byte-identical, including seed. Their outcomes varied on 7 of 19 tasks. The remaining 12 tasks were stable: ten fixed passes and two fixed failures. Later model sampling, tool outputs, timing, and trajectories were uncontrolled execution variation, but actor-visible task, model, seed field, tools, workspace, system instructions, retained-context field, and tool protocol were the same. Three of these six labels are also the labeled ablation comparators in Table 1, so the dispersion reference and frozen comparator are overlapping rather than independent samples.

The one-way random-effects ICC treats tasks as targets and the six named no-context labels as exchangeable single replicate executions on the observed score scale. It is a descriptive repeatability coefficient, and the three-target ActiveGraph value is especially unstable. The designizing diagnostic is $(1.96 + 0.842)\widehat{SE}$, using a two-sided nominal $\alpha = 0.05$, 80% power, and the within-task variance across the six equivalent labels. It is a continuous normal approximation for a paired mean contrast. On the binary families it does not represent an attainable observed score increment. The three-task ActiveGraph value is illustrative only and must not be used to interpret the continuous-score spread as a stable power estimate. Neither planning quantity is a decision threshold.

All discriminative variation in the no-context probe was concentrated in seven tasks that changed outcome under equivalent requests. Stable tasks still provide one-sided opportunities: fixed passes can expose regressions and fixed failures can expose gains. Minimal’s evolved failure on `scikit-learn_scikit-learn-13124`, against

Table 1: Frozen held-out outcomes. Intervals are nominal descriptive task-resampling intervals; $[0, 0]$ intervals are degenerate observed contrasts, not population certainty.

Family	Configuration	Evolved	Labeled ablation	Evolved-ablation [interval]	Sham	Evolved-sham
SWE (10)	Workspace	8/10	8/10	0/10 $[0, 0]$ pp	6/10	+2/10
	Minimal	7/10	6/10	+1/10 $[-20, +40]$ pp	7/10	0/10
	Hybrid	8/10	7/10	+1/10 $[0, +30]$ pp	8/10	0/10
Terminal (6)	Workspace	5/6	4/6	+1/6 $[0, +50]$ pp	3/6	+2/6
	Minimal	3/6	5/6	-2/6 $[-66.7, 0]$ pp	4/6	-1/6
	Hybrid	4/6	4/6	0/6 $[0, 0]$ pp	4/6	0/6
ActiveGraph (3)	Workspace	82.7%	100.0%	-17.3 $[-52, 0]$ pp	96.0%	-13.3 pp
	Minimal	96.7%	88.0%	+8.7 $[-10, +36]$ pp	100.0%	-3.3 pp
	Hybrid	82.7%	92.7%	-10.0 $[-40, +10]$ pp	82.7%	0.0 pp

Figure 2. Expression profiles are multidimensional

Minimal’s 1.35% storage reach coexists with 100% semantic reach; no single byte ratio is sufficient.

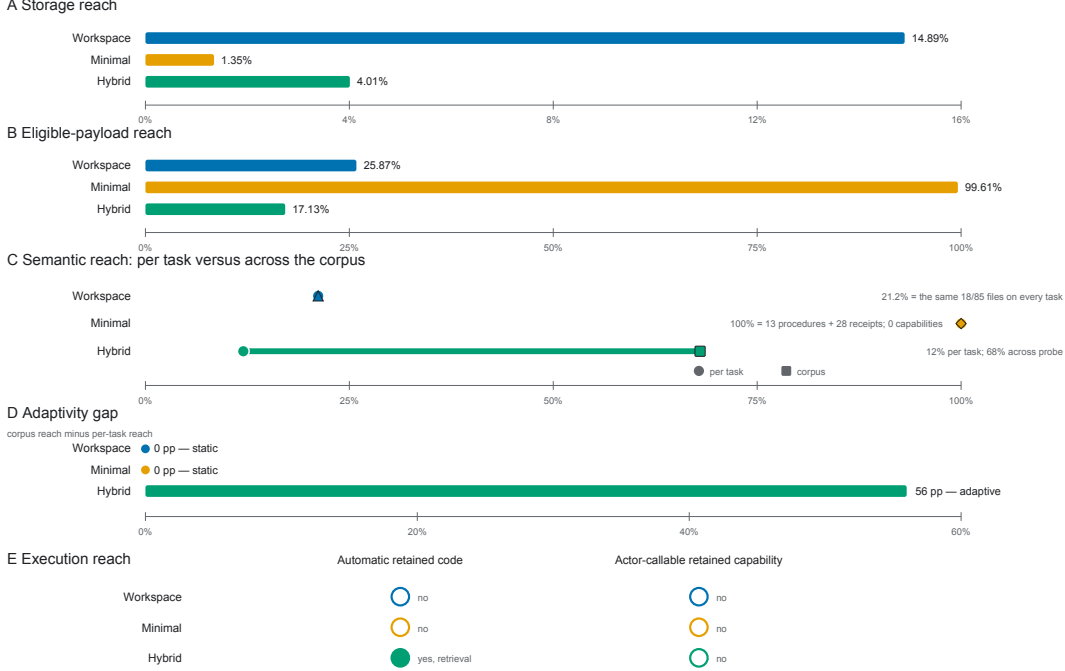


Figure 2: Expression is a profile. Five axes are reported: storage reach, eligible-payload reach, semantic coverage with per-task and probe-union submeasures, selection adaptivity, and execution reach with automatic and actor-visible submeasures. Percentages are within-substrate coverage measures and do not compare quantities of useful information. Minimal’s 41 declared units are 13 procedures and 28 receipts; none is executable. Hybrid conditions top-three selection on task text, exposing three of 25 lessons per task and 17 across the probe. Its Pack runs automatically in the adapter and supplies no on-demand actor-visible invocation handle.

Table 2: Retained state after 28 development tasks.

Substrate	Accepted after		Final retained product
	Pass	Fail	
Workspace	13	12	429,731-byte workspace
Minimal	13	0	13 procedures, 0 capabilities, 28 receipts
Hybrid	15	10	25 lessons; 62,506-byte Pack

six equivalent passes, was the only one of 57 evolved configuration-task results outside an exact task-specific equivalent range. Under a tie-aware conditional exchangeability calculation, the expected count is 12/7 (1.71): 1/7 among 36 cases on stable control tasks and 11/7 among 21 cases on variable control tasks. The observed count

was one and therefore supplies no positive chance detection. Method and family breakdown are recorded in `data/generated/outside-range-exchangeability.json`.

The maximum of the 15 absolute pairwise label-mean gaps is reported for each family. The 15 values are dependent functions of six labels, and the labels are not randomized evolution replications. The maximum is a descriptive dispersion reference rather than a significance threshold.

The sham comparison exposes a second control problem. Workspace SWE evolved minus sham was +2/10 tasks (+20.0 points) because the evolved and ablation arms each solved 8/10 tasks while the opaque sham solved 6/10; equivalent no-context labels on the same tasks also ranged from 6/10 to 8/10. On Terminal, Workspace

Table 3: Expression profile. Coverage ratios are within-substrate and do not compare useful information. Workspace counts a bounded partial-file prefix once and reports completeness separately.

Substrate	Storage	Eligible	Semantic: task; union	Selection; expansion	Execution: automatic; handles
Workspace	14.9%	25.9%	18/85 (21.2%); 18/85	static; 0 pp	none; 0, retained tools hidden
Minimal	1.35%	99.6%	41/41 (100%); 41/41	static; 0 pp	none; N/A, 0 executable units
Hybrid	4.0% mean	17.1% mean	3/25 (12%); 17/25 (68%)	task-conditioned; 56 pp turnover	automatic Pack query; 0 handles

Table 4: Equivalent duplicate-execution dispersion and planning diagnostics. Neither planning quantity is a decision threshold.

Family	Tasks	Tasks variable across six labels	ICC(1,1)	Maximum pair gap	Continuous-approx. planning
SWE	10	3	0.72	2/10 (20.0 pp)	30.7 pp
Terminal	6	2	0.66	1/6 (16.7 pp)	45.1 pp
ActiveGraph	3	2	0.31, unstable	21.3 pp	28.7 pp

evolved minus sham was +2/6 tasks (+33.3 points) because evolved solved 5/6, ablation solved 4/6, and sham solved 3/6; equivalent labels on those tasks ranged from 4/6 to 5/6. The SWE sham family score fell within the equivalent-label range, at 6/10 against 6/10 to 8/10. The Terminal sham fell below all six equivalent family means, at 3/6 against 4/6 to 5/6. With six dependent draws, a seventh observation below the observed minimum is consistent with interference and does not establish it. The sham consumed approximately 1.88 to 2.43 times as many incremental first-call tokens per byte as evolved context, depending on configuration. Its context bytes exactly matched evolved context for Workspace and Minimal. For Hybrid, the 6,093-byte sham was 56% to 58% of the mean evolved context across families. Byte matching therefore controlled artifact size while leaving tokenization, semantic load, and interference uncontrolled for Workspace and Minimal; Hybrid also retained a substantial byte-size mismatch.

7 Mechanism audits

7.1 Failure-memory stickiness

Hybrid retained 25 lessons, including 15 from passing development attempts and ten from failed attempts. Across 19 held-out tasks, the top-three retriever filled 57 slots. Failure-derived lessons occupied 37/57 slots (64.9%) despite forming 10/25 of the corpus (40%). Two failed Pylint lessons occupied 22/57 slots (38.6%). Only 17/25 lessons were ever retrieved, and 5/57 slots shared a repository prefix with the held-out task.

The implemented ranker counted set overlap between task words and an index containing suite, task identifier, title, scope, and trigger terms. Failure-derived lessons had broader indexed vocabularies. The raw enrichment, 37/57 retrieval slots from 10/25 failure-derived lessons, is the total association. Indexed vocabulary correlated with retrieval count within passing lessons ($\rho = 0.743, n = 15$) and failed lessons ($\rho = 0.763, n = 10$). In the descriptive regression `retrievals ~ indexed vocabulary + failure flag`, vocabulary had a coefficient of 0.194 retrievals per additional term and the adjusted failure coefficient was -0.093 retrievals. A 50,000-draw descriptive

within-status label-permutation diagnostic shuffled vocabulary values within pass/fail strata, refit the vocabulary coefficient, and counted absolute coefficients at least as large as observed with a plus-one Monte Carlo correction. It gave $p_{MC} \approx 0.0045$. This conditional reference requires exchangeability of vocabulary labels within status; the 25 lessons form one fixed corpus and compete for 57 retrieval slots, so the value is neither causal nor population inference.

Offline re-ranking the same prompts changed the selection pattern without new model calls. Failure slots fell from 37/57 under raw overlap to 33/57 under Jaccard, 27/57 under binary cosine, and 28/57 under BM25. BM25 surfaced 22/25 lessons and reduced the top-two share from 22/57 to 10/57. These diagnostics are consistent with vocabulary-mediated selection in the implemented overlap ranker. They do not identify whether failure status has additional effects outside that path and do not show that alternative retrieval improves downstream task scores.

7.2 Capacity saturation

Hybrid Pack source grew from 7,053 to 62,506 bytes. Generation 25 first exceeded the 64,000-byte policy and succeeded after one repair. Every candidate in generations 26 through 28 remained over the cap after four author or repair requests; the closest final candidate was 65,928 bytes. Mean authoring requests rose from 1.0 over the first five generations to 3.2 over the last five, and mean authoring cost rose from \$0.249 to \$2.114.

The policy rejection and retained-state plateau directly establish policy-cap saturation under this append-oriented design. Task identity and difficulty are confounded with generation order, so the cost trend is descriptive. The result does not establish a general limitation of ActiveGraph or executable memory.

7.3 Update-proposal taxonomy

A frozen codebook was applied to all 84 generation-level reflection proposals. Each coder saw opaque shuffled excerpts without architecture, task, sequence, score, acceptance, cost, or later retrieval metadata. Sol and Terra completed separately executed, outcome-blind coding passes using different model families. Sol also gener-

Figure 3. Evolved outcomes versus task-specific duplicate-execution clouds

Six byte-identical no-context requests define each gray cloud; Minimal/scikit-learn is the only evolved result outside a cloud.

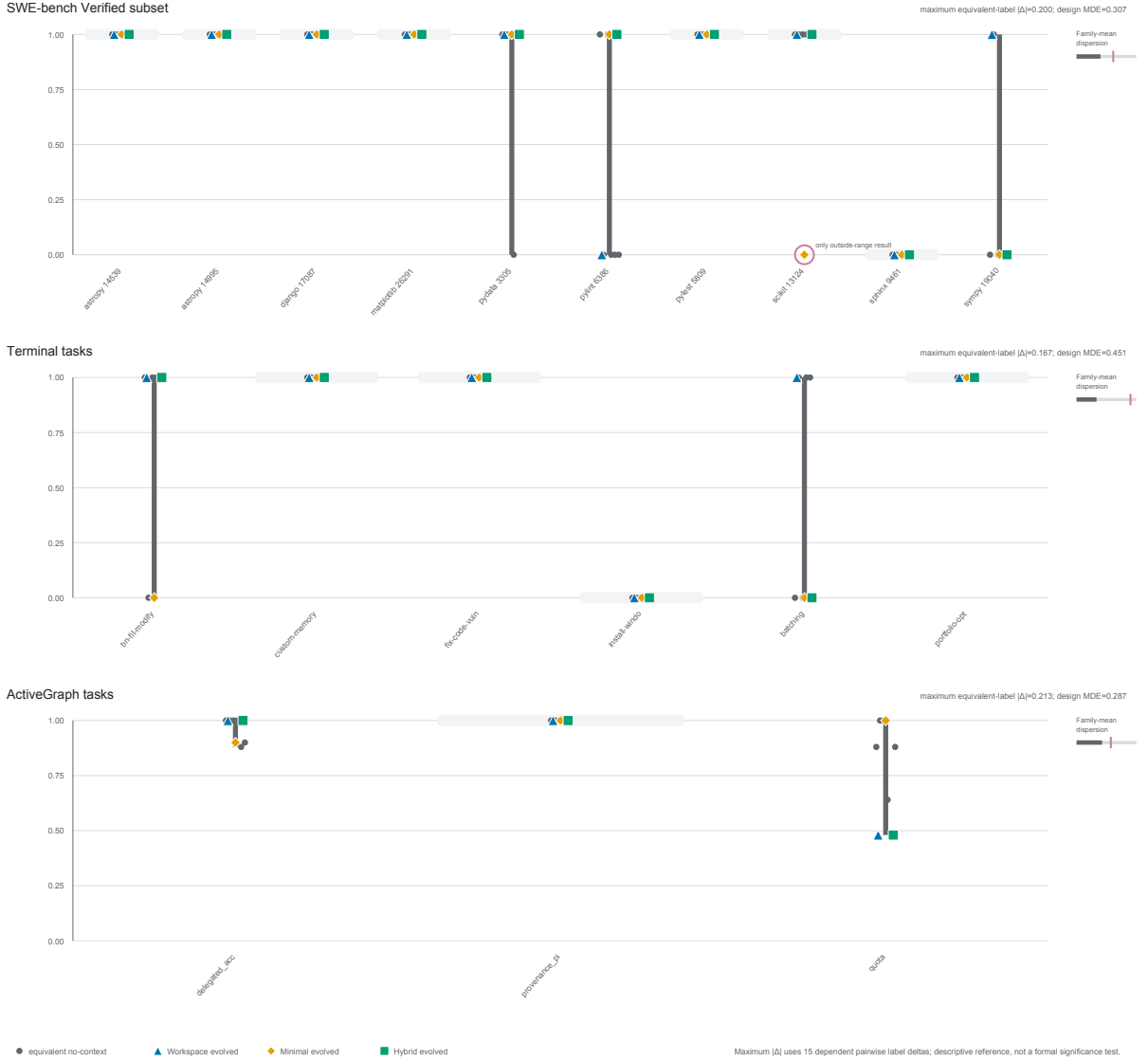


Figure 3: Task-level initial-request duplicate spread. For each of 19 tasks, gray points show six no-context outcomes generated from byte-identical initial model requests; colored marks show the three evolved outcomes. Later trajectory inputs may diverge. Seven tasks varied under the duplicate executions and 12 were stable. Minimal/scikit-learn is the only evolved configuration-task outcome outside the observed duplicate-execution range. The bar is the maximum of 15 dependent absolute pairwise label-mean gaps and is descriptive rather than a formal significance threshold. Neither the bar nor the planning diagnostic is a decision threshold.

ated the proposals it later coded; Terra did not. Original labels were hashed before comparison and no disagreement was adjudicated.

Both coders labeled 84/84 proposals as natural-language, append-only, nonexecutable, operational guidance with no specified activation. They agreed on failure specificity for 83/84 proposals and validation strategy for 82/84. These replicated labels support a stable description of proposal form.

Interpretive fields were less stable and remain exploratory. Their complete agreement results and coder

marginals appear in Appendix E.

The shared reflector therefore produced a homogeneous representational genre. Configuration-specific gates selected different retained mixtures: Minimal accepted 13 success-derived proposals and no failure-derived proposals; Workspace accepted 25 proposals including 12 after failures; Hybrid accepted 25 including ten after failures. Human inter-rater reliability remains pending. The 84/84 form agreement is a cross-model consistency check, not independent human reliability. The analysis describes common update proposals rather than every na-

Figure 4. Failure-memory stickiness is a vocabulary-coverage mechanism

Failures are 40% of memory but 64.9% of retrievals; vocabulary predicts selection within both outcome groups.

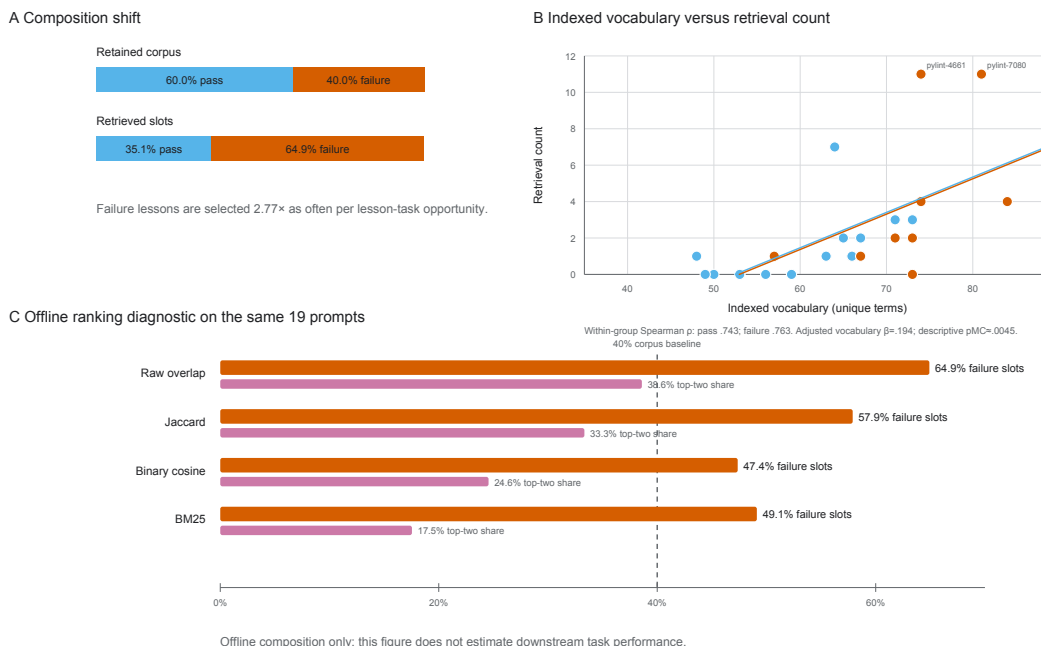


Figure 4: Failure-memory stickiness and lexical coverage. Hybrid’s retained corpus was 40% failure-derived, while failures occupied 37/57 retrieval slots (64.9%). Retrieval count increased with indexed vocabulary within both success and failure groups. Because failure status plausibly acts through indexed vocabulary breadth, the adjusted regression is a mechanism diagnostic rather than an attribution of signal away from failure. It does not establish an effect on held-out score.

tive Workspace edit or Pack source change.

8 Audited cases

Case selection followed a frozen rule that required a complete chain:

retained update -> expression -> plausible mechanism -> later behavior

The rule also required comparison with the exact task-specific range among equivalent duplicates.

Apparent success. Workspace passed a held-out SymPy task. Its retained SymPy lesson concerned a different mechanism, and a no-context run generated essentially the same fix. Equivalent outcomes on the task ranged from failure to pass. The receipts establish a pass and reject the learned-transfer story.

Apparent failure. Hybrid scored 0.48 on quota scheduler after retrieving mechanistically skewed lessons. Six equivalent no-context runs on the same task ranged from 0.48 to 1.00. The evolved score equaled the observed minimum. The receipts establish retrieval skew and reject attribution of this outcome to that skew.

Outside-range regression. Minimal failed a scikit-learn task that all six equivalent no-context runs passed. The retained-context trajectory changed an algorithm and rewrote an established expected-output test. The abla-

tion made a smaller fix and passed the restored official test. This is consistent with harmful behavioral mediation in that trace. Across all 57 configuration-task cases, however, one outside-range result was observed against a tie-aware conditional expectation of 1.71, so the case is not a population or chance detection. The trace does not identify one retained sentence as the cause.

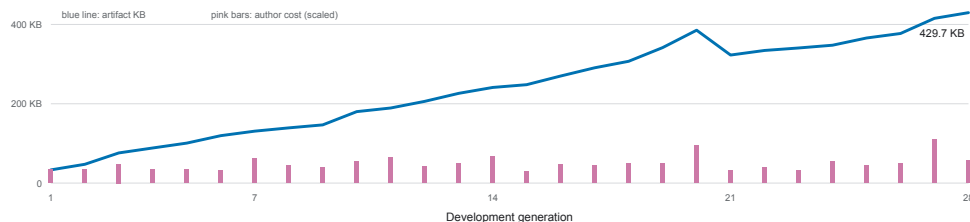
Capacity ceiling. Hybrid generation 25 succeeded after a repair near the policy cap. Generations 26 through 28 produced four over-cap candidates each. This case establishes a direct mechanism from append-only accumulation to failed retention.

Together, the cases show why receipts must be permitted to invalidate stories in both directions. They also show why score, mechanism, and attribution should be reported separately.

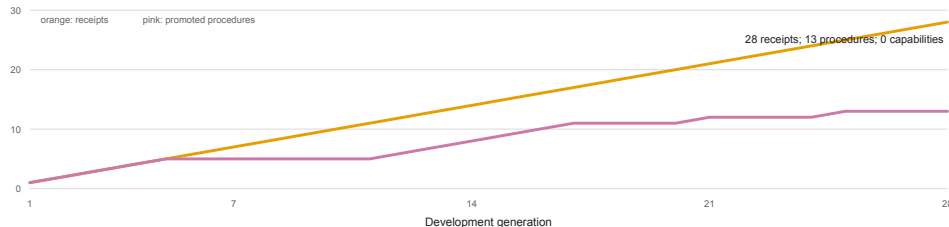
Figure 5. Accumulation, selective promotion, and saturation

All substrates retained state; only Hybrid encountered an explicit capacity ceiling.

A Workspace: artifact growth and authoring cost



B Minimal: success-gated procedures and universal receipts



C Hybrid: Pack source reaches the 64 KB policy ceiling

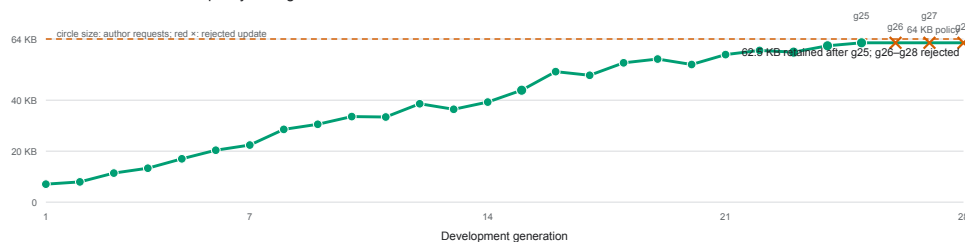


Figure 5: Accumulation, rising effort, and saturation. Panel A shows Workspace artifact growth and authoring cost; panel B shows Minimal’s universal receipts and success-gated procedures; panel C shows Hybrid Pack growth and rejected updates at the 64,000-byte source policy. Hybrid generation 25 succeeded after repair and generations 26 through 28 produced no within-cap candidate. The cost trend is descriptive because generation order is confounded with task identity and difficulty.

9 Related work

Self-improving agent systems measure different portions of the five-construct chain. STOP reports repeated recursive improver optimization and held-out task utilities. Gödel Agent [5] evaluates runtime self-referential modification with tool and policy ablations. SICA optimizes a coding-agent source tree through a reported primary lineage. Darwin Gödel Machine uses a branching archive and reports transfer across benchmarks, models, and languages. Huxley-Gödel Machine explicitly measures descendant productivity through clade-metaproductivity.

Our study complements these capability results by instrumenting retained-state attribution. Table 6 separates unevolved baselines, mechanism interventions, independent complete evolution runs, versioned provenance, immutable exposure receipts, and recursive-improvement operationalization. No design reports every listed feature. This study lacks independent evolution-lineage replication and does not identify descendant productivity. It adds sham state, initial-request duplicates, and immutable exposure receipts to a persistent post-curriculum comparison.

Reflexion [6] shows that episodic linguistic feedback can alter later trials; AI Agents That Matter [7] moti-

vates cost, holdout, and attribution controls. Stochasticity in Agentic Evaluations [8] motivates consistency analysis; PACE [9] studies noisy acceptance. Useful Memories Become Faulty When Continuously Updated by LLMs [10] contrasts episodes with consolidation; DeltaMem [11] studies residual-tree memory. Together they motivate measurement of retained-state mechanisms.

The closest conceptual precedents come from three adjacent traditions. Memory evaluation distinguishes stored, retrieved, selected, and utilized material; causal mediation separates intervention, mediator, trajectory, and outcome; systems provenance and observability bind events without by themselves identifying causal effects. Our residual contribution is to unify these distinctions for persistent self-modification, keep descendant productivity as a separate recursive endpoint, profile expression across heterogeneous retained substrates, and bind configuration-specific claims to receipts and actor-visible controls. The framework is a domain-specific evidentiary synthesis rather than a claim that each component is unprecedented.

Table 6: Related-work evidence features. “Yes” means that the cited primary paper reports the feature. “Partial” identifies a nearby but weaker design. “NR”

means not reported and does not imply that the authors could not implement it. Cells were re-audited against the linked primary paper versions on 2026-07-23. The strict receipt column requires immutable binding among retained state, evaluation exposure, trajectory, and outcome.

Table 6: Author-classified related-work evidence features based on primary papers read on 2026-07-23. NR means not reported and is not independently verified here.

Panel A: substrate, transfer, and interventions									
System	Retained surface		Held-out transfer		Initial baseline	Mechanism	intervention	Sham state	
STOP [1]	Python prover/scaffold	im-	Yes: independent LPN and five new utilities		Seed improver	Nonrecursive and alternative improvers		NR	
Gödel Agent [5]	Runtime logic/actions		Partial: held-out splits		Initial policy and manual baselines	Initial-tool ablations		NR	
SICA [2]	Coding-agent source tree		NR: utility reused in primary run		Initial agent	NR		NR	
DGM [3]	Coding-agent branching archive		Cross-benchmark, model, language		Initial agent and manual baselines	Without improvement/open exploration	self-	NR	
HGM [4]	Coding-agent search tree		Benchmark and model transfer		Shared initial agent	Matched SICA/DGM selection	NR		
This study	Workspace; procedures/receipts; Pack	proce-	19 disjoint curriculum tasks	post-	Cold floor and initial harness	Partial: state hash preserved; actor-visible ablation collapses with cold	Opaque sham; mismatch reported		

Panel B: replication, provenance, and recursive-improvement evidence									
System	Complete runs	evolution	Identical request	initial-duplicates	Versioned eage/provenance	lin-	Immutable sure/trajectory ceipts	expo-re-	Descendant productivity
STOP	Five GPT-4; 25 smaller-model in key experiments		NR		NR		NR		NR
Gödel Agent	Six cycles per task		NR		NR		NR		NR
SICA	NR: one primary lineage		NR		Yes: archive of agents/results		NR		NR
DGM	Three complete runs in stability analysis		NR		Traceable archive tree		NR		NR
HGM	NR for repeated complete searches		NR		Retained search tree		NR		Yes: clade-
This study	No: one lineage per con-figuration		Six labels per task		Immutable tions/hashes	genera-	State, exposure, request, trajectory, grader, adjudication		No: not identified

10 Discussion

The frozen null does not establish that persistent self-modification cannot improve agents. It establishes that three retained configurations, under one common interface and one lineage each, did not meet the specified uplift criterion. The mechanism audit explains why a score-only summary would discard the most informative evidence.

First, native update breadth and evaluation-time expression are separate design dimensions. Workspace retained arbitrary files, Minimal retained procedures and receipts, and Hybrid retained executable Pack source. The actor received truncated static text, complete static text, or task-conditioned text. It received no callable retained capability. Evaluation therefore compared native substrates after a consequential common transformation.

Second, proposal, acceptance, retrieval, and expression form a pipeline. The common reflector produced one representational genre across all 84 proposals. Acceptance gates created different epistemic mixtures. Workspace and Hybrid retained failure-derived proposals that Minimal excluded. Hybrid’s lexical retriever then concentrated exposure on broad-vocabulary failure lessons. Each mechanism belongs inside the system boundary because each changes what a later actor can use.

Third, control labels can collapse or confound contrasts. Cold and the nominal state-hidden ablation preserved different provenance but delivered the same actor-visible intervention, so the frozen single-ablation comparison is one draw from the no-context distribution. Pooled-six and leave-one-ablation-out sensitivities preserve the null. The opaque sham matched evolved bytes for Workspace and Minimal but not Hybrid, and it differed in tokenization and semantic load. Equivalent duplicates varied despite identical first requests. A positive evolved-minus-sham contrast can therefore combine useful expression, possible sham interference, and ordinary outcome spread. The control ladder makes these alternatives explicit and suggests the next diagnostic comparison.

Fourth, append-only retention carries a measurable maintenance burden. Hybrid reached a hard cap; Workspace and Minimal accumulated without consolidation. This observation motivates consolidation, pruning, and replacement as first-class update operations. It does not establish which operation would improve held-out performance.

The instruments are directly applicable in principle to persistent agents whose retained-state inventory, eligibility rule, adapter emissions, and actor-visible actions can be frozen and inspected. Systems without byte-copy semantics should declare a computable proxy or mark the byte axes not applicable. Equivalent duplicates can be retained whenever multiple labels issue byte-identical initial requests under a common execution contract. The control ladder can be selected according to the intended causal claim. The update-proposal taxonomy can be applied to shared reflectors while native state changes are analyzed separately. Receipts can bind retained artifacts,

exposures, trajectories, graders, and adjudication.

11 Limitations

The study completed one development lineage per configuration. The six equivalent labels expose outcome variability and do not replace independent evolution replications. All configurations shared one model and outer agent, which may dominate configuration differences. The held-out families contained ten, six, and three tasks. ActiveGraph uncertainty estimates are especially unstable.

Cold and cold ablation were actor-visible-equivalent. Although cold ablation validated a lineage state hash, it provided no lineage-derived input, tool, workspace, instruction, or later protocol difference. The frozen primary contrast is therefore not a separable matched-lineage intervention. Pooled and leave-one-ablation-out sensitivities use the available equivalent controls but do not create independent evolution-lineage replication.

The common interface constrained native expression and exposed no retained interactive capability. This limits architecture ranking and creates the expression bottleneck examined in the paper. A general auditability-expressivity relationship requires an intervention that varies the interface while holding retained state fixed.

The sham targeted byte matching rather than token or semantic matching, and Hybrid retained a substantial byte-size mismatch. It remains useful as an interference control and cannot identify equal-compute semantic effects. Duplicate-label pairwise gaps are dependent and post-hoc. The task-level ranges are descriptive.

The update-proposal taxonomy used two language-model coders, one of which also generated the proposals it coded. Exact agreement on representational form is a cross-model consistency check and does not substitute for independent human coding. Interpretive labels showed lower exact agreement and remain exploratory. Configuration masking may be incomplete because proposal form can reveal substrate.

The retrieval analysis identifies lexical selection mechanics and does not establish causal harm to score. Offline re-ranking produced no new behavioral outcomes. No retrieval-harm experiment or consolidation intervention is part of this study.

The local ActiveGraph benchmark and the Hybrid Pack share author-built technology. Sealed checks and an observed result unfavorable to Hybrid reduce a simple directional-bias concern but do not remove the structural conflict.

12 Conclusion

All three recorded configurations retained persistent changes. Their evaluation interface expressed those changes mainly as text, their controls exhibited substantial outcome spread and left interference uncontrolled, and none met the frozen uplift criterion. The resulting

null separates five claims that are often compressed into one: retention, expression, behavioral mediation, task improvement, and recursive improvement.

The central lesson is operational. Treat reflection templates, acceptance gates, retrievers, adapters, execution paths, and controls as components of the self-improving system. Measure each link with receipts and matched interventions. In the studied system, the interface and acceptance mechanism substantially determine which retained modifications can be admitted, expressed, and evaluated.

Artifact availability. The version-controlled release includes the paper, generated tables, frozen coding artifacts, figures, audit scripts, and a self-verifying manifest. Complete trace archive DOI pending its separate disclosure scan and deposit.

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A Adjudication sensitivity

The raw run contained 215 immediately valid rows and 13 rows requiring policy adjudication. The final dataset retained all 228 attempts. Categories were six official SWE task timeouts, six Terminal agent-budget timeouts with completed zero verifiers, and one ActiveGraph snapshot contradiction resolved by a sealed grader-only retry of the hash-identical submission.

Eight of nine frozen `evolved - cold ablation` contrasts are identical when raw invalid rows are scored zero and when the frozen adjudication decisions are applied. The exception is Hybrid ActiveGraph: the contrast moves from -26.0 points under raw-invalid-as-zero to -10.0 points after the hash-identical quota-scheduler regrade. It remains negative and the frozen conclusion is unchanged. A raw complete-case analysis gives +7.3 points for this cell because it drops the evolved invalid row. Complete-case contrasts are reported only as sensitivity analyses because dropping bounded failures changes the estimand and can favor arms with more invalid outcomes.

The study’s single adjudication judgment moved a score in the evolved arm’s favor, and the frozen conclusion remained null.

Machine-readable table: `data/generated/adjudication-sensitivity.csv`

B Timeout taxonomy

Timeouts remain outcomes under the bounded-agent estimand. The taxonomy separates agent budget exhaustion, official evaluator limits, and infrastructure contradiction.

C Success-conditional efficiency

For the binary SWE and Terminal families, the companion table reports mean cost and wall time among successful attempts for every architecture and arm: `data/generated/success-conditional-efficiency.csv`.

These quantities describe the resource profile of observed successes. They do not estimate causal efficiency because conditioning on success selects different tasks and trajectories across arms. Unconditional costs remain the primary resource comparison.

D Provenance

The study specification is `research/local_study.json`, SHA-256 `8313d21971452704893200d2022718feca985142ae32fe7d16932d34a0e0d581`, with frozen status at study commit `559843894578141dfbd39b7bae2246db76e57bde` dated `2026-07-17T18:30:38-07:00`. The compact release does not retain an independently witnessed timestamp for the first development call. The Round 1 manuscript package was commit `8bdf502cc08be968b21bbe5ebf177d49879a0fb`.

The provenance manifest records byte counts and SHA-256 hashes for the paper package’s frozen analytical datasets and reports, adjudication ledger, post-hoc metrics, masked coding packet, both original coder label files, agreement report, and manuscript: `provenance-manifest.json`.

The adjudication ledger separately hashes each changed row and its supporting evidence. Both original taxonomy-label files were frozen before adjudication.

E Cross-model coding agreement

The table reports exact agreement, Cohen’s kappa, and Gwet’s AC1 for two separately executed, outcome-blind model-coding passes using different model families. Sol generated the 84 proposals it later coded; Terra supplied the other pass. The analysis is therefore a cross-model consistency check rather than independent human reliability. Kappa is undefined when both coders use one category for every item because the marginals have zero variance. AC1 is a supplemental prevalence-robust coefficient computed with the full pre-specified category vocabulary. Its value of 1.000 for invariant fields is mechanical and supplies no evidence beyond 84/84 exact agreement.

For primary update target, Sol assigned 76 procedure and eight validation-asset labels, while Terra assigned all 84 to procedure. The resulting confusion is 76 procedure/procedure and eight validation-asset/procedure cases, which explains 90.5% exact agreement with kappa 0.000. Complete coder marginals and disagreements are retained in `data/taxonomy/coder-agreement.json`.

No coefficient interval is reported because these are two fixed model-coding passes used as a prompt and model-sensitivity check rather than sampled human raters. Interpretive fields remain exploratory.

F Actor-visible control-collapse sensitivity

Source inspection confirms that cold and cold ablation use the same task, model, seed field, tools, workspace, system instructions, empty retained-context field, and later actor-visible tool protocol. Cold ablation validates the lineage state hash before suppressing its contents; that provenance difference never reaches the actor. The frozen `evolved - cold ablation` contrast is therefore retained as a decision record, not described as a separable lineage intervention.

The pooled-six and leave-the-matched-ablation-out calculations use only the committed compact score table. Across Workspace, Minimal, and Hybrid, the pooled-six deltas are +6.7, -3.3, and +6.7 points for SWE; +11.1, -22.2, and -5.6 for Terminal; and -9.9, +4.1, and -9.9 for ActiveGraph. The corresponding leave-one-out deltas are +8.0, -6.0, +6.0; +10.0, -20.0, -6.7; and -8.4, +3.2, -9.9 points. The frozen no-uplift decision is unchanged.

Table 7: Timeout and adjudication taxonomy.

Category	Count	Treatment	Interpretation
Official SWE task timeout	6	Scored zero	Official evaluation exceeded the frozen 3,600-second task limit.
Terminal agent-budget timeout	6	Scored zero	Agent exhausted 1,800 seconds and the completed verifier returned zero.
ActiveGraph snapshot regrade	1	Hash-identical grader-only retry	Original grader contradicted a receipt-bound parsable submission; no model action was rerun.

Table 8: Cross-model coding consistency. Sol generated the proposals it coded; undefined kappa indicates zero marginal variance.

Field	Exact agreement	Cohen’s kappa	Gwet’s AC1
Primary update target	76/84 (90.5%)	0.000	0.904
Secondary update target	47/84 (56.0%)	-0.081	0.543
Representational form	84/84 (100%)	undefined	1.000
Transfer scope	63/84 (75.0%)	0.470	0.717
Abstraction level	59/84 (70.2%)	0.529	0.647
Evidence grounding	59/84 (70.2%)	0.428	0.669
Failure specificity	83/84 (98.8%)	0.661	0.988
Validation strategy	82/84 (97.6%)	0.592	0.976
Consolidation operation	84/84 (100%)	undefined	1.000
Executable status	84/84 (100%)	undefined	1.000
Anticipated activation	84/84 (100%)	undefined	1.000
Counterfactual actionability	84/84 (100%)	undefined	1.000
Novelty relative to prior state	84/84 (100%)	undefined	1.000
Confidence	63/84 (75.0%)	0.393	0.687

A tie-aware conditional exchangeability calculation pools each evolved score with its six actor-visible-equivalent controls and uniformly relabels the focal score. Only a unique minimum or maximum is strictly outside the other six. Across 57 configuration-task cases, the expected outside-range count is 12/7 (1.71) and the observed count is one. The expected counts are 1/7 among 36 cases on stable control tasks and 11/7 among 21 cases on variable control tasks. Machine-readable outputs are `data/generated/equivalent-control-sensitivity.csv` and `data/generated/outside-range-exchangeability.json`.